

Assume for now

$$\begin{cases} (\sin x)' = \cos x \\ (\cos x)' = -\sin x \\ (e^x)' = e^x \\ (\ln x)' = \frac{1}{x} \end{cases}$$

Ex. Prove $(x^n)' = nx^{n-1}$

Pf Using induction

Base Case If $n=0 \Rightarrow x^n = C$

$$(x^0)' = 0 = 0x^{n-1}$$

Induction step Assume $(x^n)' = nx^{n-1}$

$$\text{WTS: } (x^{n+1})' = (n+1)x^{(n+1)-1} = (n+1)x^n$$

$$x^{n+1} = x^n \cdot x$$

Apply Product Rule

$$(x^{n+1})' = (x^n)' \cdot x + x^n \cdot (x)'$$

$$= nx^{n-1} \cdot x + x^n \cdot 1$$

$$= nx^n + x^n$$

$$= (n+1)x^n$$

Ex ① Differentiate

$$f(x) = [(\sin x)(\cos x)](x^2)$$

$$\text{By product rule, } f'(x) = [\cos^2 x - \sin^2 x] x^2 + 2x[(\sin x)(\cos x)]$$

② Differentiate $f(x) = \sin(x^2 + \sin x)$

$$f'(x) = \cos(x^2 + \sin x) \cdot (2x + \cos x) \quad \text{By chain Rule}$$

Leibniz Notation We also $f'(x) = \frac{df}{dx}$
 $f'(a) = \left. \frac{df}{dx} \right|_{x=a}$

The 2nd derivatives are written as $\frac{d(\frac{df}{dx})}{dx} = \frac{d^2f}{(dx)^2}$
3rd derivatives is $\frac{d^3f}{(dx)^3}$, etc.

The chain Rule in Leibniz Notation is $\frac{d(f \circ g)}{dx} = \frac{df}{du} \cdot \frac{du}{dx}$

The derivative of $f \circ g$ at pt a : $\left. \frac{d(f \circ g)}{dx} \right|_{x=a} = \left. \frac{df}{du} \right|_{u=g(a)} \cdot \left. \frac{du}{dx} \right|_{x=a}$

Spivak Ch.10 #22

The number a is called a double root of f if $f(x) = (x-a)^2 g(x)$.
Show f has a double root a

$\Leftrightarrow f$ has a root at a + f' has a root at a .

Solu ($\Rightarrow$) If f has a double root, then $f(x) = (x-a)^2 g(x)$

$$\text{so } f'(x) = 2(x-a)g(x) + (x-a)^2 g'(x)$$

$$f'(a) = 2 \cdot 0 \cdot g(a) + 0 \cdot g'(a)$$

$$= 0$$

$\therefore f'$ has a root at a .

($\Leftarrow$) Suppose $f(a) = f'(a) = 0$

Write $f(x) = (x-a)g(x)$

Differentiate: $f'(x) = g(x) + (x-a)g'(x)$

Since f' has a root at a :

$$f'(a) = g(a) = 0$$

$\Rightarrow g$ has a root at a

$$\Rightarrow g(x) = (x-a)h(x)$$

$$\text{So } f(x) = (x-a)^2 h(x)$$



Spivak Ch. 9 #14

Let $f(x) = \begin{cases} x^2, & x \in \mathbb{Q} \\ 0 & \text{otherwise} \end{cases}$, show $f(x)$ is differentiable at $x=0$.

Solve By definition, the derivative of f at 0 (if f exists) is

$$\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

$$\text{Now, WTS: } \lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$$

Let $\varepsilon > 0$. want to find $\delta > 0$ s.t. whenever $0 < |h| < \delta$

$$\text{we have } \left| \frac{f(h)}{h} - 0 \right| < \varepsilon$$

Strategy: Separate δ 's for $h \in \mathbb{Q}$, $h \in \mathbb{R} \setminus \mathbb{Q}$. then take the min of the 2.

Case 1 $h \in \mathbb{R} \setminus \mathbb{Q}$

$$\left| \frac{f(h)}{h} - 0 \right| = |0 - 0| = 0 < \varepsilon$$

any δ works.

Case 2 $h \in \mathbb{Q}$, then $\left| \frac{f(h)}{h} - 0 \right| = \left| \frac{h^2}{h} - 0 \right|$

$= |h| < \varepsilon$ * If $f(h) = h$ for $h \in \mathbb{Q}$, this is $|1-0|$
so limit DNE

Take $\delta = \varepsilon$ then it works

Then for $\delta = \varepsilon$, $\left| \frac{f(h)}{h} - 0 \right| < \varepsilon \quad \forall 0 < |h| < \delta$

i.e. $\lim_{h \rightarrow 0} \frac{f(h)}{h} = f'(0) = 0$

So the derivative exists, & it is 0.



Spivak Ch. 9 #17

Let $0 < \beta < 1$. Show that if $|f(x)| \geq |x|^\beta \quad \forall x$, $f(0) = 0$, then f is not differentiable at 0.

Pf By definition, if f were diff at 0, then $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$ exists
 $= \lim_{h \rightarrow 0} \frac{f(h)}{h}$

Note that $\lim_{h \rightarrow 0} \left| \frac{f(h)}{h} \right| \geq \lim_{h \rightarrow 0} \frac{|h|^\beta}{|h|}$
 $= \lim_{h \rightarrow 0} |h|^{\beta-1}$

$\lim_{h \rightarrow 0} \frac{1}{|h|^{1-\beta}} \xrightarrow[\beta < 1]{h \rightarrow 0} \infty$



Claim: If $\lim_{x \rightarrow a} g(x) = l$, then $\lim_{x \rightarrow a} |g(x)| = l$

Pf. Let $\varepsilon > 0$, want $\delta > 0$ s.t.

whenever $0 < |x-a| < \delta$, then $||g(x)| - |l|| < \varepsilon$

We know $\lim_{x \rightarrow a} g(x) = l$

So for the given $\varepsilon > 0$, $\exists \delta' > 0$ s.t. whenever $0 < |x - a| < \delta'$, we have $|g(x) - l| < \varepsilon$

By Δ ineq, we have

- $|g(x)| - |l| \leq |g(x) - l|$
- $|l| - |g(x)| \leq |l - g(x)|$

$$||g(x)| - |l|| < |g(x) - l| < \varepsilon$$

$$\text{if } 0 < |x - a| < \delta$$

If $\lim_{h \rightarrow 0} \frac{f(h)}{h}$ exist

$$\text{then } \lim_{h \rightarrow 0} \left| \frac{f(h)}{h} \right| = |l| \rightarrow \infty$$

contradiction

